TONY POULAKOS | INDEPENDENT TECHNICAL PORTFOLIO

Retail Demand & Supply Intelligence

Executed Databricks prototype and Power BI reporting | Independent LEGO Group-inspired case study

From synthetic source data to validated reporting

A reproducible retail analytics demonstration connecting sales, demand, inventory, and replenishment to a shared reporting model. Built to show practical data engineering, metric definition, and business interpretation.

51,840
store-product-days

18 / 18
quality gates passed

6
reporting model tables

3
Power BI report pages

BUSINESS QUESTIONS

How are sales performing across regions?

Where does inventory constrain simulated demand?

How well does a transparent forecast baseline predict demand?

VERIFIED SIMULATION RESULTS

$8.70M synthetic sales revenue
92.43% consumer demand fill
94.46% daily stock availability
88.92% supplier unit fill

12 fictional stores, 24 fictional products, and 180 days: Jan 1-Jun 29, 2026.

DECISION RELEVANCE

West contributes **30.1%** of revenue. South has the highest modeled stockout exposure: **$193.0K**.

The baseline underforecasts demand by **6.04%**, with **53.10% WAPE** at store-product-day grain.

Scope: All operational data and outcomes are synthetic. The $707,795 modeled exposure (see Executive Overview tab) is unmet demand multiplied by selling price, not demonstrated savings. This independent prototype is not an official LEGO Group system or an enterprise production deployment.

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Engineering Approach & Evidence

Python | PySpark | SQL | Delta Lake | Power Query | DAX

SOURCE
Deterministic simulator

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BRONZE
Raw Delta snapshots

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SILVER
Validated, conformed data

>

GOLD
Facts, dimensions, marts

>

POWER BI
Snapshot import + DAX

PROCESSING & DATA QUALITY

Bronze stores raw synthetic inputs and ingestion metadata. Silver quarantines three deliberately invalid sales rows and removes five duplicate deliveries. Gold materializes SQL reporting tables over Delta Lake.

Eighteen fail-fast checks cover completeness, unique keys, foreign keys, quantity bounds, forecast coverage, inventory conservation, stock continuity, receipts, and revenue reconciliation.

Run evidence: status SUCCEEDED; 51,840 clean daily rows; 8,064 forecast rows. The export records 124.69 seconds to KPI capture; this is a single demo run, not a performance benchmark.

REPORTING MODEL & FORECAST

Three dimensions (date, store, product) filter three facts (daily activity, forecast, replenishment) through nine active, single-direction relationships. Eighteen DAX measures power the three report pages.

A fixed-origin, 28-day holdout evaluates a prior 56-day weekday-mean forecast. Training excludes the holdout. Actual demand is directly observed in the simulator; real lost demand would require estimation.

Delivery boundary: Power BI imports verified Gold CSV snapshots; it is not a live Databricks connection. The notebook performs repeatable full refreshes with audit logging. Scheduled jobs and incremental production ingestion are not configured.

Evidence chain: executed Gold CSVs + runtime_results.json + quality_checks.csv; independently recalculated KPIs; populated Power BI Desktop report and PDF export.
Run ID: 6e0a1217-0e7c-4879-bde4-4665ad250a5b

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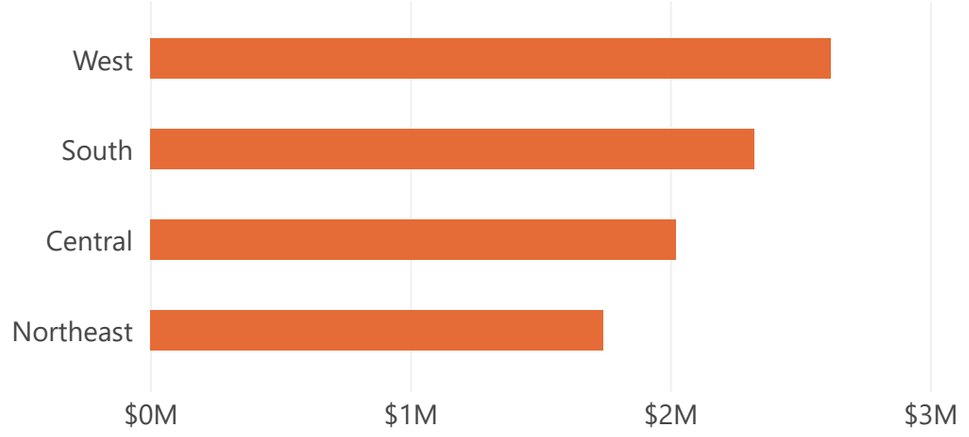
Retail Demand & Supply Intelligence — Executive Overview

Synthetic LEGO Group retail demonstration | January–June 2026 | Fictional stores and products

$8.70M	92.43%	94.46%	$707.80K
Revenue USD	Consumer Demand Fill Rate	Daily Availability Rate	Modeled Stockout Exposure USD

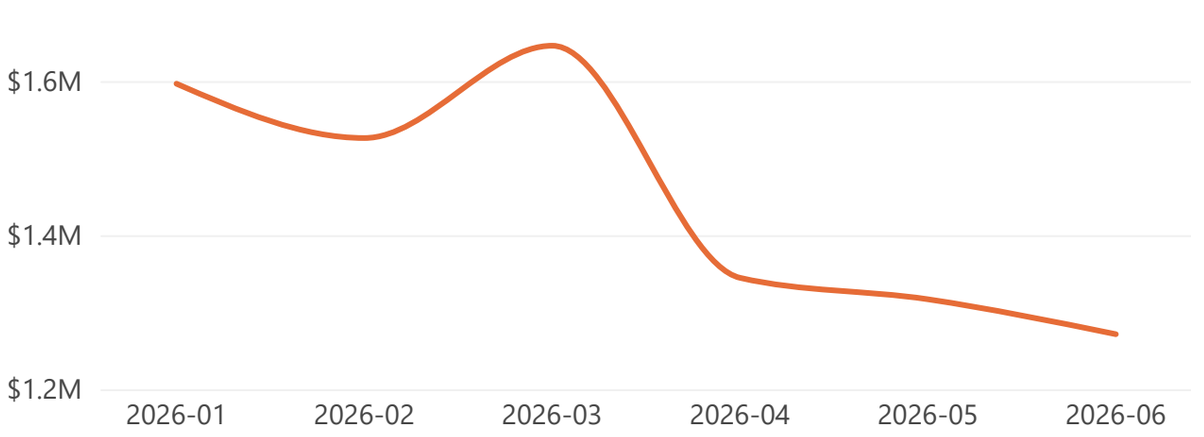
Sales revenue by region

Revenue USD by Region



Monthly revenue | June through June 29

Revenue USD by year-month



18 / 18 pipeline quality gates passed | Revenue, keys, forecast coverage, and inventory reconciled

Databricks → Delta Lake → Power BI | 12 fictional stores · 24 products · 51,840 store-product-days

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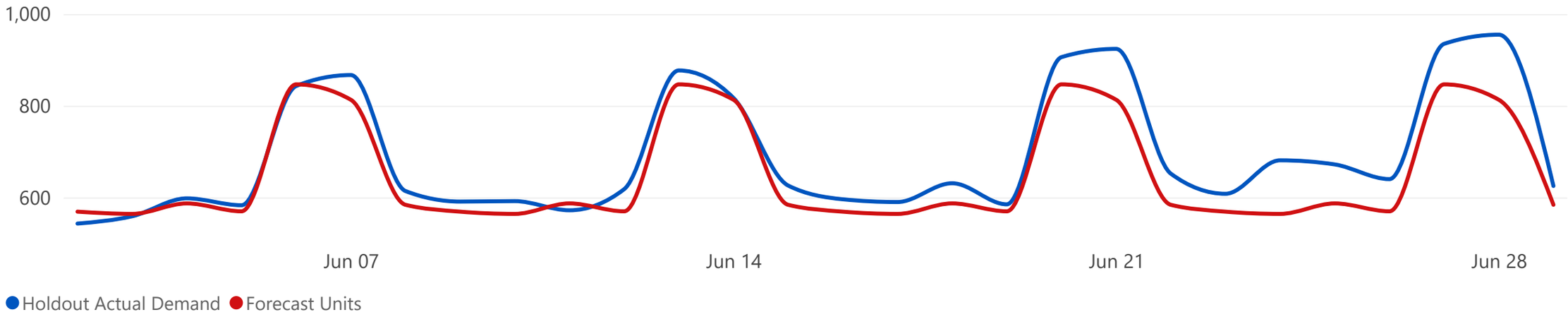
Retail Demand & Supply Intelligence — Demand Forecast

Synthetic LEGO Group retail demonstration | January–June 2026 | Fictional stores and products

53.10%	-6.04%	28	8,064
Forecast WAPE	Forecast Bias	Holdout Days	Forecast Records

28-day holdout: simulated demand vs. baseline forecast | Units

Holdout Actual Demand and Forecast Units by date



Baseline: prior 56-day weekday mean | WAPE at store-product-day grain; network totals mask local error

Databricks → Delta Lake → Power BI | 12 fictional stores · 24 products · 51,840 store-product-days

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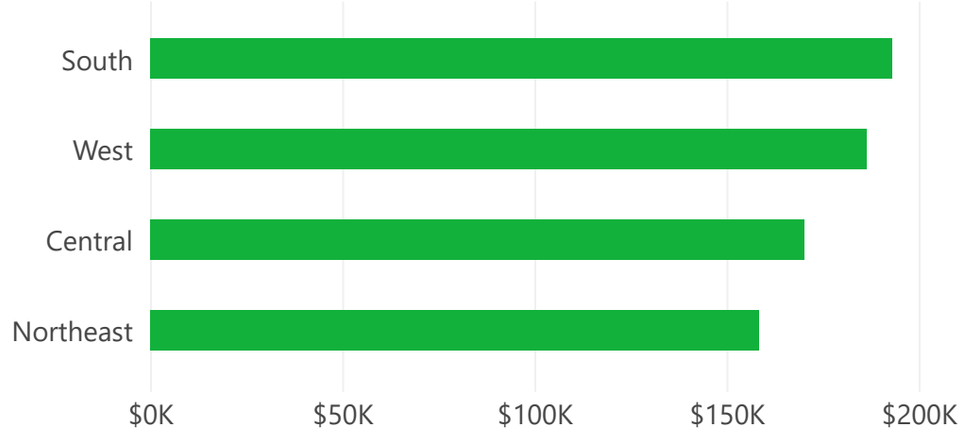
Retail Demand & Supply Intelligence — Inventory & Supply

Synthetic LEGO Group retail demonstration | January–June 2026 | Fictional stores and products

88.92%	7.75%	10,089	2,548
Supplier Fill Rate	Stockout Day Rate	Lost Demand Units	Closing Inventory Units

Modeled stockout exposure by region

Modeled Stockout Exposure USD by Region



Inventory exposure by fictional theme

Theme	Lost Demand Units	Modeled Stockout Exposure USD
City-inspired	2,544	$176,614.56
Creative bricks	2,409	$168,545.91
Space-inspired	2,703	$188,393.47
Vehicles-inspired	2,433	$174,241.17
Total	**10,089**	**$707,795.11**

Exposure is modeled unmet demand × price; not demonstrated savings. Closing inventory is the period-end snapshot.

Databricks → Delta Lake → Power BI | 12 fictional stores · 24 products · 51,840 store-product-days